RAJALAKSHMI ENGINEERING COLLEGE

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D2

Started on	Friday, 7 August 2026, 9:39 PM
State	Finished
Completed on	Friday, 7 August 2026, 9:46 PM
Time taken	7 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
void function(int n)
{
    int c= 0;
    c=c+1;
    for(int i=n/2; i<n; i++){
        c=c+2;


        for(int j=1; j<n; j = 2 * j){
            c=c+ 2;


            for(int k=1; k<n; k = k * 2){
               c=c+2;
            }
        }
    }
    c=c+1;
    printf("%d\n",c);
}
int main(){
    int n;
    scanf("%d",&n);
    function(n);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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